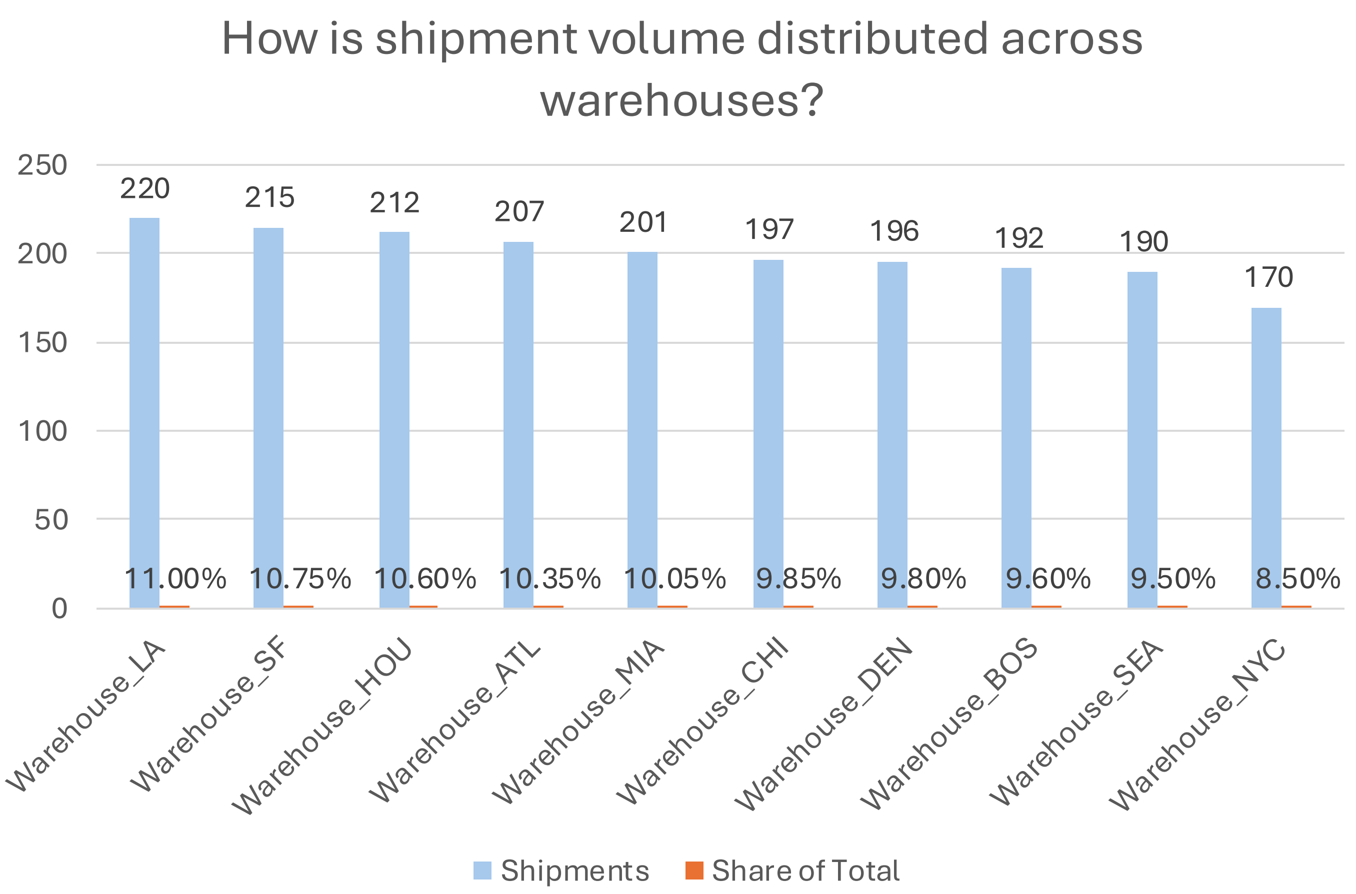
2023 Logistics Performance Overview

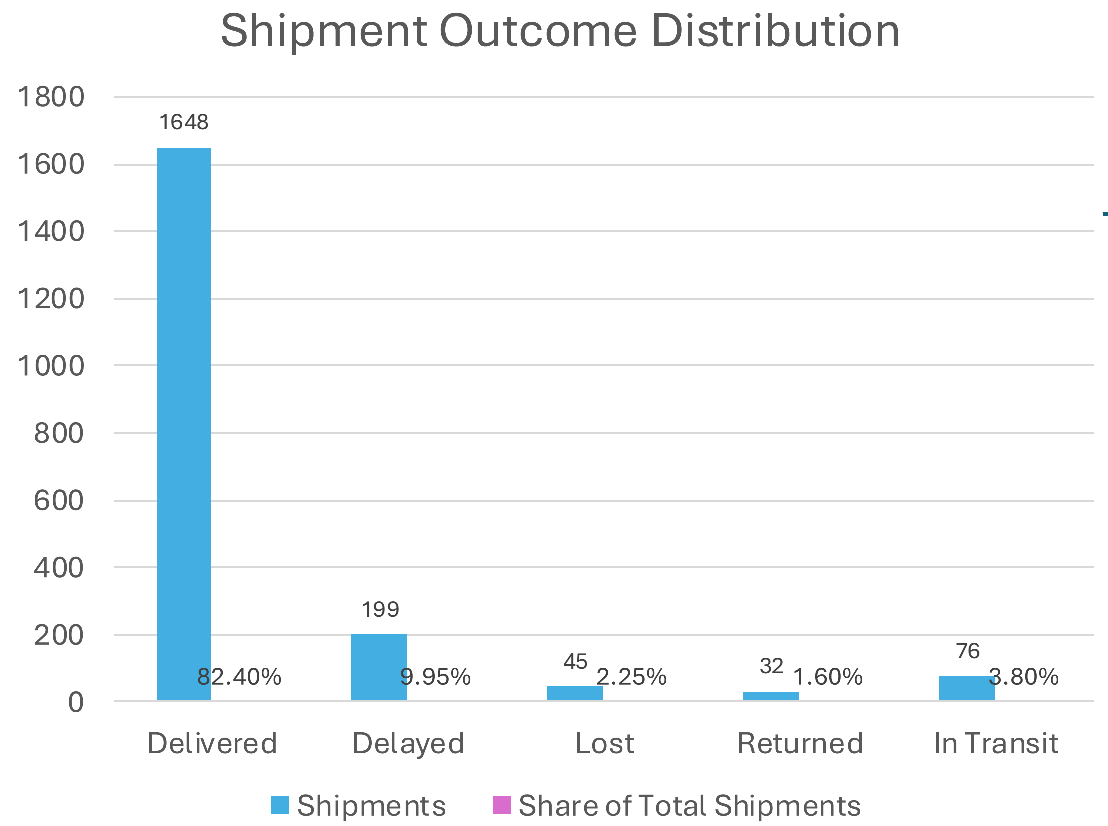
What is the overall size of the logistics network?

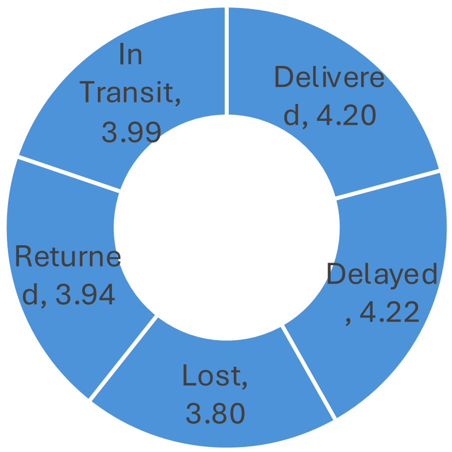
Metric	Value
Total Shipments	2,000
Total Transportation Cost	$401,911.57
Total Distance Covered	2.55M miles
Average Shipment Cost	$205.16
Average Transit Time	4.18 days



Overall Delivery Performance

- What percentage of shipments were successfully delivered?
- What percentage experienced issues?





Delivery time is almost the same across all shipment statuses.

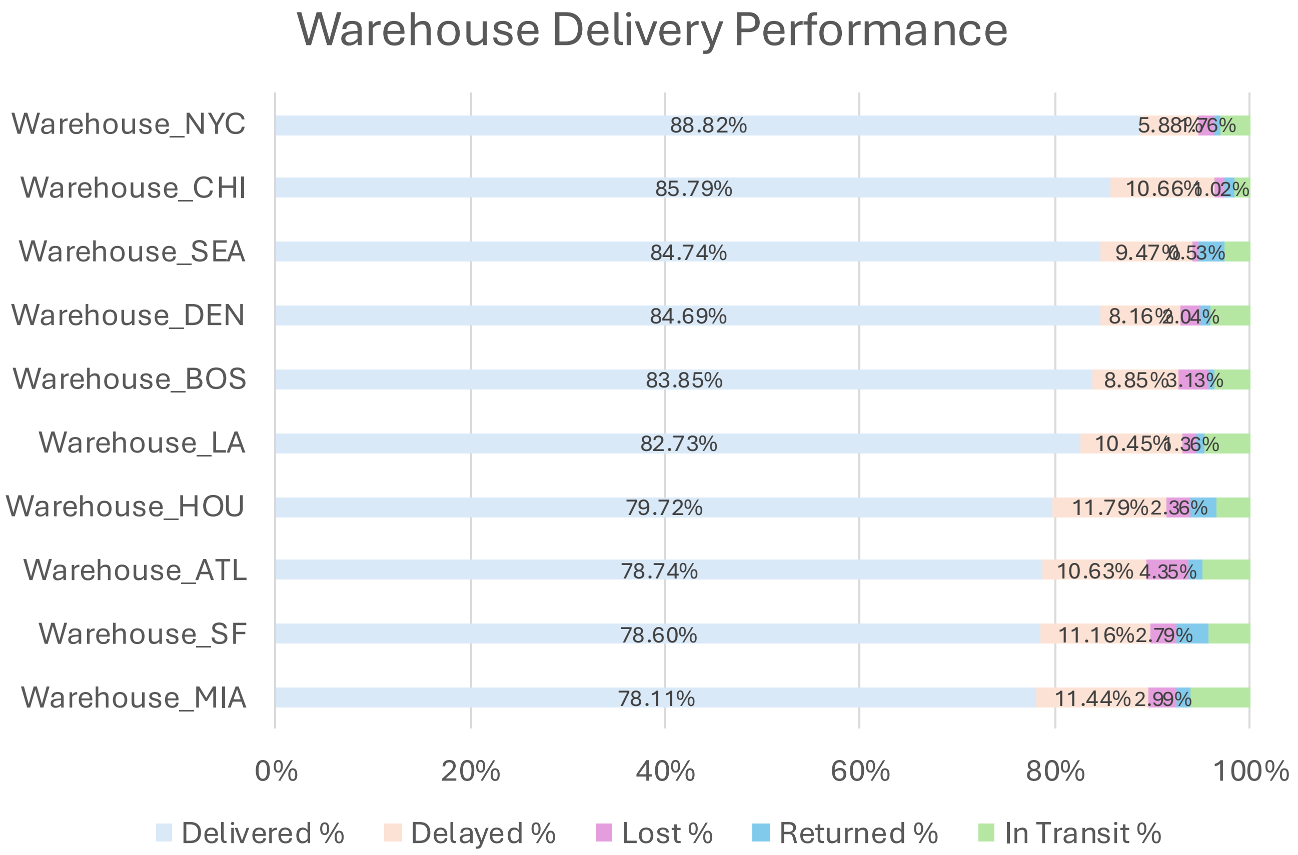
Delay issues are likely concentrated in specific parts of the network rather than caused by overall delivery speed.

Delivery Status	Average Shipping Cost	Total Shipping Cost
Delivered	$208.24	335,687.78
Delayed	$201.66	39,122.86
Lost	$175.32	7,889.19
Returned	$196.36	6,283.64
In Transit	$170.11	12,928.10

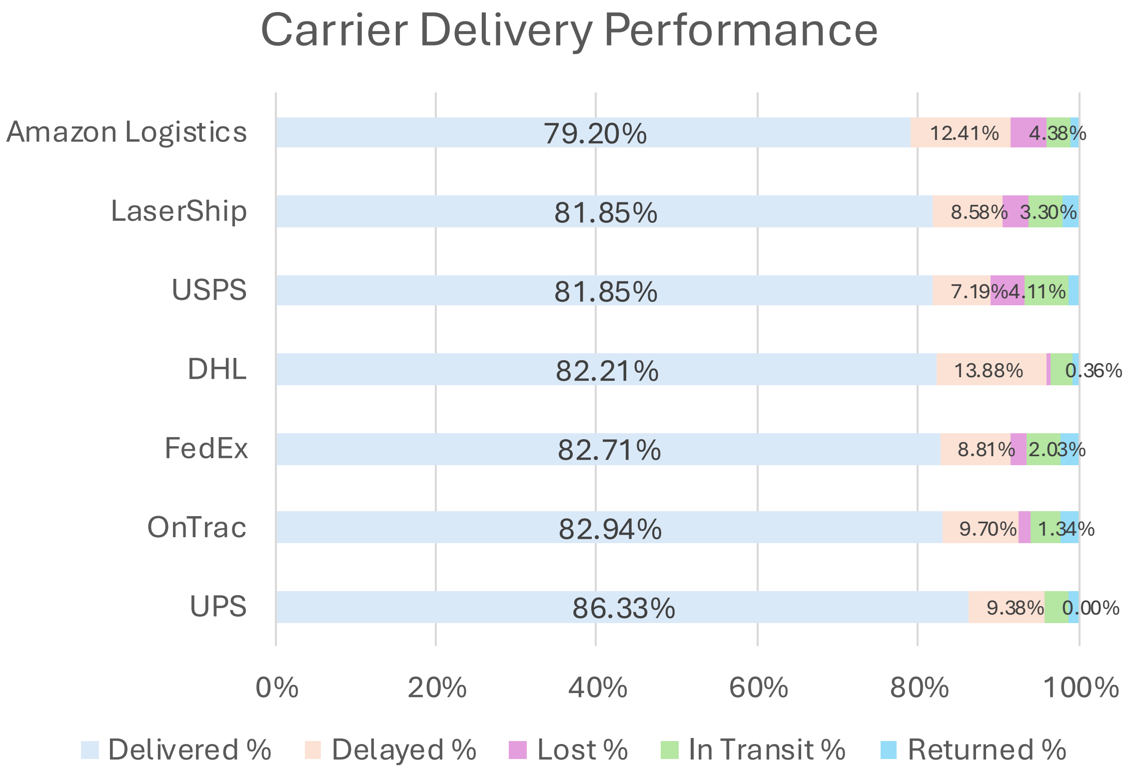
Lost shipments account for 2.25% of volume but resulted in $7.9K of transportation costs without completed delivery.

Priority: identify where lost shipments occur most often (warehouse and carrier) to reduce avoidable losses.

Where Are Delivery Problems Occurring?



MIA, SF, ATL, and HOU require operational review due to higher rates of delayed, lost, and unresolved shipments.



- Amazon Logistics requires review due to the lowest delivery success rate and highest loss rate among carriers.

Which warehouse - carrier combinations create the highest delivery risk?

Warehouse	Carrier	Shipments	Delivered %	Delayed %	Lost %
Warehouse_MIA	USPS	25	68.00%	12.00%	12.00%
Warehouse_HOU	Amazon Logistics	29	68.97%	20.69%	6.90%
Warehouse_LA	FedEx	26	69.23%	19.23%	3.85%
Warehouse_HOU	LaserShip	33	69.70%	9.09%	6.06%
Warehouse_ATL	Amazon Logistics	33	69.70%	18.18%	9.09%
Warehouse_ATL	DHL	27	70.37%	18.52%	0.00%
Warehouse_BOS	LaserShip	31	70.97%	12.90%	9.68%
Warehouse_SF	FedEx	28	71.43%	0.00%	3.57%
Warehouse_MIA	OnTrac	28	71.43%	7.14%	0.00%
Warehouse_LA	USPS	32	71.88%	12.50%	3.13%
Warehouse_ATL	LaserShip	26	73.08%	11.54%	7.69%
Warehouse_NYC	Amazon Logistics	15	73.33%	6.67%	6.67%
Warehouse_BOS	Amazon Logistics	23	73.91%	17.39%	4.35%
Warehouse_SF	Amazon Logistics	32	75.00%	15.63%	9.38%

- **MIA-USPS should be reviewed first due to delivery failures: 12% of shipments were lost, creating direct cost exposure.**
- **HOU-Amazon Logistics requires investigation because 1 in 5 shipments were delayed (20.7%), increasing customer waiting time.**
- **ATL-Amazon Logistics shows both delay and loss issues (18.2% delayed, 9.1% lost), making this combination a priority for corrective action.**

Warehouse	Carrier	Shipments	Delivered %	Delayed %	Lost %
Warehouse_LA	UPS	30	90.00%	10.00%	0.00%
Warehouse_CHI	FedEx	30	90.00%	10.00%	0.00%
Warehouse_ATL	USPS	31	90.32%	0.00%	6.45%
Warehouse_NYC	LaserShip	23	91.30%	8.70%	0.00%
Warehouse_NYC	DHL	23	91.30%	8.70%	0.00%
Warehouse_BOS	UPS	26	92.31%	7.69%	0.00%
Warehouse_LA	Amazon Logistics	30	93.33%	3.33%	0.00%
Warehouse_NYC	FedEx	31	93.55%	6.45%	0.00%
Warehouse_NYC	UPS	18	94.44%	5.56%	0.00%
Warehouse_DEN	UPS	18	94.44%	5.56%	0.00%
Warehouse_BOS	FedEx	22	95.45%	4.55%	0.00%
Warehouse_CHI	LaserShip	28	96.43%	0.00%	0.00%

- **CHI–LaserShip shows the best performance: 96.4% of shipments delivered with no delays or losses.**
- **Top-performing locations can provide examples of processes or practices that may help improve lower-performing areas.**

Shipment Characteristics and Delivery Issues

Are delivery issues associated with shipment characteristics?

Delivery Status	Average Shipping Cost	Avg Distance	Avg Weight
Delivered	$208.24	1289.13	31.01
Delayed	$201.66	1263.50	28.23
Lost	$175.32	1133.71	26.60
Returned	$196.36	1254.44	24.67
In Transit	$170.11	1113.80	22.35

Weight → Similar across statuses → No clear relationship with delivery issues.

Distance → Similar across statuses → Longer routes do not show more issues.

Shipping Cost → Lower for lost shipments → Delivery issues are not concentrated in higher-cost shipments.

How can delivery performance be improved?

Audit warehouse operations:

! Review order processing workflows.
Improve picking and packing procedures.
Reduce handling errors.
Ensure SLA compliance.

Strengthen carrier management:

! Review carrier contracts and service levels.
Reallocate shipment volume from underperforming carriers where appropriate.
Establish regular carrier performance reviews.

Improve warehouse–carrier coordination:

! Review workflows for high-risk warehouse-carrier combinations.
Standardize operating procedures across locations.
Reassign carrier routes where operationally feasible.

Reduce shipment losses:

! Strengthen package tracking and scanning procedures.
Improve inventory control during handoffs.
Introduce additional quality checks before shipment.

Replicate best practices:

! Adopt successful processes from top-performing warehouses.
Standardize operating procedures across the network.
Share operational best practices across teams.

How is transportation cost distributed across the network?

Which warehouses account for the largest transportation spending?

Warehouse_Name	Total Cost Per Warehouse	Share of Total Cost	Avg Cost
Warehouse_LA	49,586.73	11.00%	229.57
Warehouse_HOU	48,155.31	10.60%	231.52
Warehouse_SF	42,647.23	10.75%	200.22
Warehouse_MIA	40,526.09	10.05%	203.65
Warehouse_ATL	40,484.36	10.35%	198.45
Warehouse_SEA	39,227.04	9.50%	209.77
Warehouse_BOS	38,344.88	9.60%	205.05
Warehouse_DEN	35,738.37	9.80%	189.09
Warehouse_CHI	35,595.22	9.85%	185.39
Warehouse_NYC	31,606.34	8.50%	192.72

LA and HOU have the highest transportation costs ($49.6K and $48.2K), representing ~22% of total warehouse spending combined.

NYC has the lowest transportation spend ($31.6K), which is ~36% lower than LA.

Which carriers account for the largest transportation spending?

Carrier	Avg Cost	Total Shipping Cost	Cost Share %
FedEx	218	64082	15.94%
LaserShip	208	61526	15.31%
DHL	222	61087	15.20%
UPS	230	57658	14.35%
OnTrac	185	54457	13.55%
Amazon Logistics	194	51774	12.88%
USPS	183	51328	12.77%

FedEx, LaserShip, and DHL represent the largest cost contributors (~46% of total transportation spend combined).

UPS has the highest average shipping cost ($230), ~26% higher than USPS ($183).

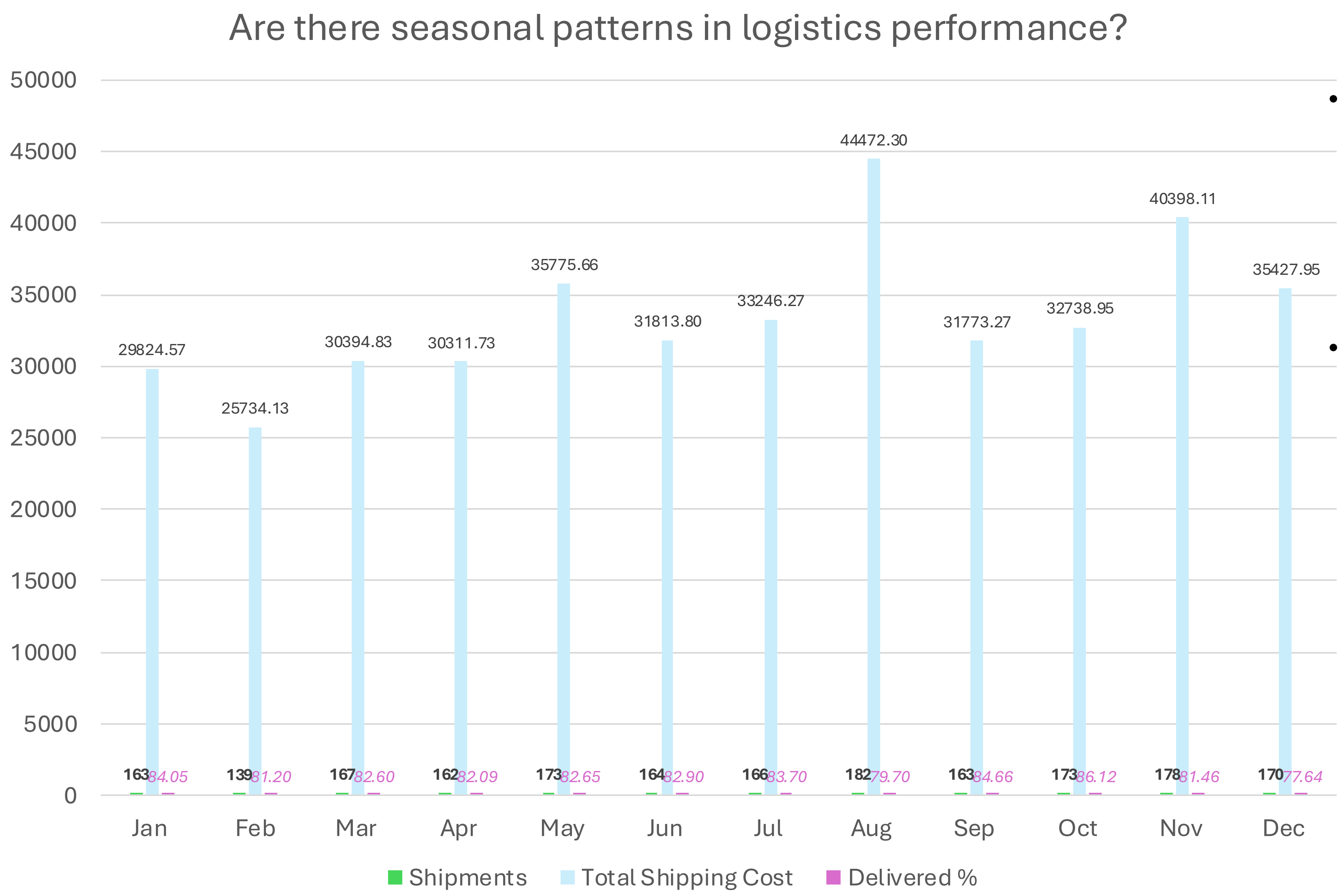
Carrier	Avg Cost	Delivered
FedEx	218	82.71%
LaserShip	208	81.85%
DHL	222	82.21%
UPS	230	86.33%
OnTrac	185	82.94%
Amazon Logistics	194	79.20%
USPS	183	81.85%

Does higher transportation cost result in better delivery performance?

UPS combines the highest average cost ($230) with the highest delivery rate (86.3%), while lower-cost Amazon Logistics shows the lowest delivery rate (79.2%).

This indicates a potential trade-off between cost and delivery performance, meaning carrier selection should consider both price and service level.

Seasonality and Operational Trends



- Higher transportation costs did not result in better delivery performance during peak months. Need to improve operational efficiency before increasing transportation spending.
- Year-end operations appear to be the most challenging period. Additional warehouse capacity and carrier planning are needed before peak periods.